

Mikołaj Mołodecki

Intern Software Engineer



mikolaj.molodecki133@gmail.com



+48 575030825



Szczecin, Poland



github.com/MiniowaPM

Summary

Highly motivated Junior Software Developer and Computer Science student with a proven track record of building real-world applications, including the published **PlanPM** app and advanced **IoT drone systems**. I bridge the gap between complex software (React, FastAPI, Flutter) and technical hardware (AI vision, 3D modeling, server admin), consistently delivering high-performance solutions in Agile environments. Driven by a passion for multimedia systems and cutting-edge rendering, I combine technical precision with a championship-level team spirit to solve the toughest engineering challenges.

Skills

Programming

JavaScript/TypeScript, Dart, C#, Python, SQL, C++, Java

Environment

- Version Control: Git, Github, GitLab
- Management Tools: YouTrack, Trello
- CI/CD Pipelines (Github Actions)
- Others: Docker

Web & App Development

- Game dev: Unity, HLSL
- Frontend: HTML, CSS, Tailwind CSS, React
- Backend: Flask, FastAPI, Spring Boot (beginner), Postman
- Mobile: Flutter, React Native

Databases

MySQL, PostgreSQL, Supabase, Appwrite

Professional Experience

06/2025 – 09/2025
Szczecin

Junior software developer intern

Global Logic

Project: IoT Project-Based Learning – World of Drones

- **Frontend Development:** Developed the frontend of the application using **React (JavaScript)** and supported the integration of the **FastAPI (Python)** backend.
- **AI & Computer Vision:** Responsible for developing and deploying an AI model based on **Nano** architecture for real-time car and license plate detection.
- **Embedded & Hardware Design:** Designed and modeled 3D protective casing for Raspberry Pi and STM32 microcontrollers using **Autodesk Fusion**. Created physical landing pads for autonomous drone operations.
- **System Architecture & Planning:** Participated in the end-to-end system design process, including hardware selection, budgeting, and cost estimation (BOM).
- **Agile Methodology:** Worked in a small, cross-functional team following **Scrum** frameworks, utilizing **Taiga** for task management and **GitLab** for version control and Jenkins for CI/CD workflows.
- **IoT Integration:** Solved complex connectivity issues between edge devices (Raspberry Pi) and backend services.

06/2022 – 07/2022
Szczecin

Altcom S.J., Szczecin

IT Support & Service Intern

- **Hardware Diagnostics & Repair:** Performed complex repairs and component replacements for both desktop PCs and laptops.
- **System Administration & Optimization:** Installed, configured, and optimized various operating systems to ensure peak performance and security.

- **Data Security & Storage Management:** Managed data backup processes and provided maintenance for **NAS servers**, ensuring full data integrity for clients.
- **Network & Server Configuration:** Configured **UNIX servers** and performed advanced data recovery procedures.
- **Technical Support (Helpdesk):** Provided direct technical support to clients, diagnosing software and hardware issues through effective communication.
- **Software Troubleshooting:** Specialized in configuring and resolving complex issues.
- **Infrastructure Maintenance:** Managed routing, DNS, and DHCP configurations as part of broader network maintenance tasks.

06/2021 – 07/2021
Szczecin

Przedsiębiorstwo Usług Specjalistycznych MVB [↗](#)

Intern

- **Assembly of Power Distribution Cabinets:** Assembled and wired power distribution units based on detailed technical documentation and electrical diagrams.
- **Technical Documentation Review:** Maintained constant communication with the design department to report discrepancies, suggest improvements, and resolve issues within the provided technical specifications.
- **Server Rack Configuration:** Assembled large-scale RACK server cabinets for industrial and commercial use.
- **Cross-departmental Collaboration:** Worked closely with the engineering and project teams to ensure all builds met strict quality and safety standards.

Selected Projects

05/2025 – Present

PlanPM [↗](#)

Tech Stack: Flutter, Supabase, Dart

- Engineered and maintained a comprehensive mobile application currently available on Google Play and the App Store, serving a live user base of students and faculty.
- Managed the full software development lifecycle (SDLC), including backend architecture, database integration (Supabase), and CI/CD pipeline configuration.
- Implemented robust error handling and monitoring, ensuring 99%+ uptime and high responsiveness through efficient state management strategies.

06/2025 – Present

Maritime Autonomous Systems Simulator

Tech Stack: C#, Unity, HLSL, Physics Simulation

- Engineered a high-fidelity maritime simulation system featuring dynamic ocean surface generation and real-time wave physics, critical for testing autonomous navigation algorithms.
- Implemented advanced shader programs (HLSL) to achieve realistic water rendering while maintaining real-time performance on simulation hardware.
- Collaborated on the integration of autonomous ship controls, focusing on low-latency state feedback and environmental simulation.

05/2026

ALEKSY – Offline Voice Assistant [↗](#)

Tech Stack: Python

- Architected a privacy-focused, fully offline voice assistant, processing the entire pipeline—from wake-word detection to speech synthesis—locally on-device.
- Optimized neural network inference for low-power hardware, ensuring stable performance without reliance on cloud APIs or external connectivity.
- Designed a modular pipeline architecture that allows for rapid language model updates and component isolation, ideal for critical, high-availability environments.

12/2025

ŁatwaUstawa [↗](#)

AI-Powered Regulatory Tracking App (Hacknation Hackathon)

Tech Stack: Python, LLM APIs, REST APIs, Mobile UI

- Co-created "ŁatwaUstawa", a mobile application that automates the tracking of new legal regulations. The project involved fetching data from government APIs and using LLM/AI models to summarize and simplify legal jargon for end-users.
- Engineered data pipelines integrating external government REST APIs to fetch real-time legislative updates.
- Implemented LLM-based processing to automatically analyze and summarize complex legal and compliance jargon into actionable insights for end-users.

Education

10/2023 – Present
Szczecin

Bachelor of Engineering

Maritime University of Szczecin [🔗](#)

- **Specialization:** Multimedia Systems Programming
- **Ongoing Research & Thesis:** *"Design and implementation of a system for generating and rendering ocean surfaces and sea waves in the Unity engine."* – Focus on real-time water physics, shader programming, and high-performance rendering.
- **Core Technical Training (Labs):**
- **Software Engineering & Management:** In-depth knowledge of software life cycle, development methodologies (Agile/Scrum), and project management.
- **Multimedia & Game Dev:** Extensive experience in game development and simulators using **Unity**, including advanced theory and real-time visualization.
- **Networking & Security:** Hands-on experience in designing local networks (VLAN, ACL) and fundamentals of **Cryptography**.
- **Data Structures & Algorithms:** Solid foundation in algorithm design and efficient data management.
- **Database Management:** Designing relational structures and optimizing queries.

09/2019 – 04/2023
Szczecin

Technical Education

Technical School of Digital Technologies [🔗](#)

- **Specialization:** Internet of Things (IoT)
- **Key Achievement:** Graduated with honors, achieving the highest scores in national professional exams (inf.02, inf.03)
- **Extracurricular Activity – School Radio Station:** * Managed technical infrastructure and sound systems for school events. Responsible for daily broadcasts, equipment maintenance, and audio signal routing. Developed teamwork and communication skills in a fast-paced environment.

Organisations

04/2025 – Present
Szczecin

IT Science Club [🔗](#)

Member

- **Industry Networking & Collaboration:** Represented the Science Club in technical conferences and workshops with leading international companies, including **Kongsberg Maritime**, **Pierce Group**, and firms from the **Automotive** sector.
- **Team Collaboration:** Working in a professional development environment using **Git** for version control and collaborating with a multi-disciplinary team of developers.

08/2024
Szczecin

EuroSciPy 2024 [🔗](#)

Volunteer & Participant

- **Active Conference Participation:** Attended technical sessions and workshops focused on the use of Python in scientific computing, data analysis, and engineering.
- **International Networking:** Engaged with global Python community experts and researchers, developing a deeper understanding of the language's ecosystem (NumPy, SciPy, Pandas).
- **Technical & Logistical Support:** Provided full-time organizational support, including technical assistance for speakers and managing infrastructure during the international event.

- **Communication:** Managed coordination tasks and attendee support entirely in English, facilitating a smooth experience for international participants.

Certificates

PL-900: Microsoft Power Platform Fundamentals

Languages

Polish
Native



English
B2/C1



Interests

Volleyball

Active member of the AZS Volleyball Club at the Maritime University of Szczecin. Represent the university in regular training sessions and official competitions. Competed in the Amateur Volleyball Leagues in Szczecin and Goleniów, as well as the semi-finals of the Polish Academic Championships. Passionate about outdoor activities, including skiing and kayaking.

Gaming

Enthusiast of both story-driven and competitive video games. Value gaming as a tool for developing strategic thinking, reflexes, and problem-solving skills.

Technology

Passionate about programming and continuous learning in the field of application development and code-based problem solving.